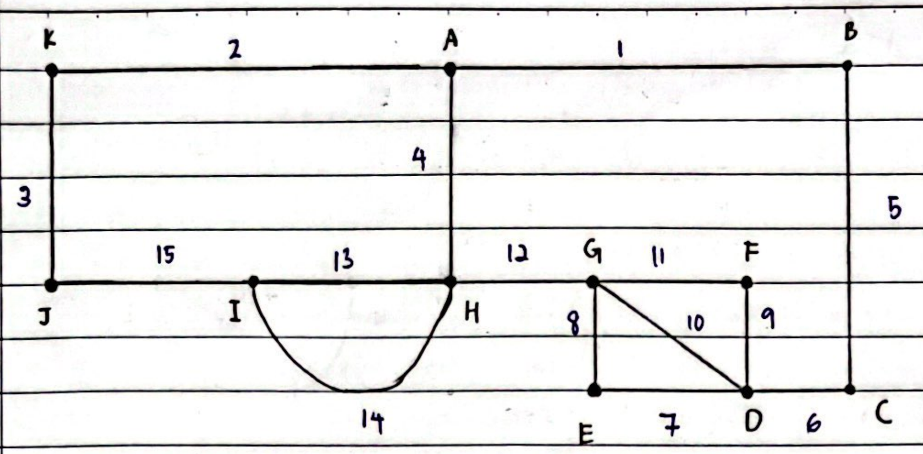
Iman Abadi Bin Mohd Nizwan (A23CS0084)
Ahmad Adib Bin A. Mazlam (A23CS0205)

No.: Date:

1.



a) (A, 1, B, 5, C, 6, D, 10, G, 11, F, 9, D, 7, E, 8, G, 12, H, 4, A, 2, K, 3, J, 15, I, 14, H, 13, I)

∴ No, the guards will not be back at the guard house at the end of the inspection

b) No it is not possible, vertex G and D needs to be visited twice in order to visit vertex F and E.

2. a)

No	S	N	L(A)	L(B)	L(C)	L(D)	L(E)	L(F)
0	{ }	{A, B, C, D, E, F}	∞	0	∞	∞	∞	∞
1	{B}	{A, C, D, E, F}	3		1	6	∞	∞
2	{B, C}	{A, D, E, F}	3			5	5	∞
3	{A, B, C}	{D, E, F}				5	5	8
4	{A, B, C, D}	{E, F}					5	8
5	{A, B, C, D, E}	{F}						7

b) (B, C, E, F), takes 7 hours

Assingment 4

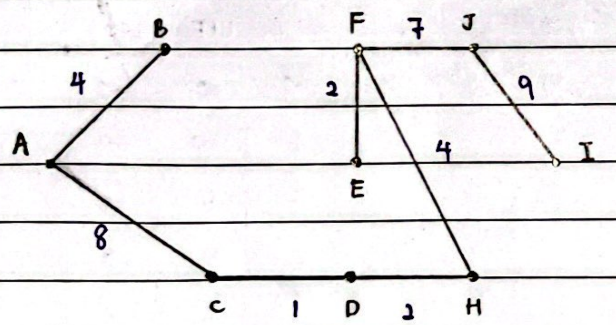
3. a) p, n, i, d, a

b) k, e, l, b, a, f, c, m, g, h, d, p, n, i, o, j

4 a) The graph is not a spanning tree

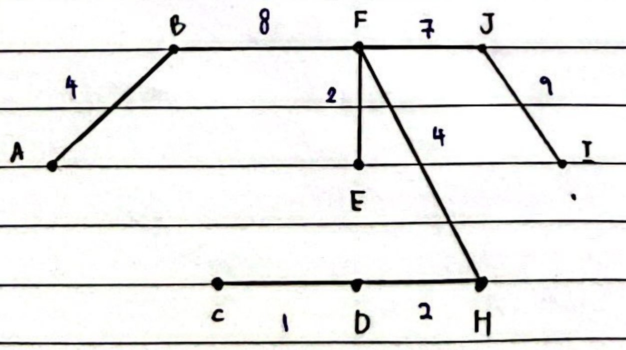
b)

Edge	Weight	Will adding edge make circuit?	Action taken	Cummulative weight of subgraph
e1 = (C,D)	1	No	Added	1
e2 = (D,H)	2	No	Added	3
e3 = (F,E)	2	No	Added	5
e4 = (A,B)	4	No	Added	9
e5 = (F,H)	4	No	Added	13
e6 = (E,D)	6	Yes	Not added	13
e7 = (E,C)	7	Yes	Not added	13
e8 = (F,I)	7	No	Added	20
e9 = (A,C)	8	No	Added	28
e10 = (B,F)	8	Yes	Not added	28
e11 = (I,J	9	No	Added	37



∴ The length is 37 metres and the cost is RM 3700.

c) Yes, by eliminating edge 9 = (A,C), edge 10 = (B,F) can be added without changing the distance and turning it into a circuit.



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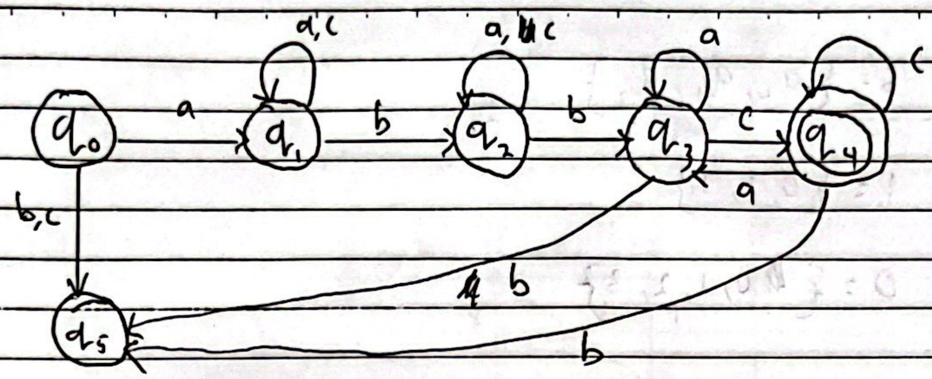
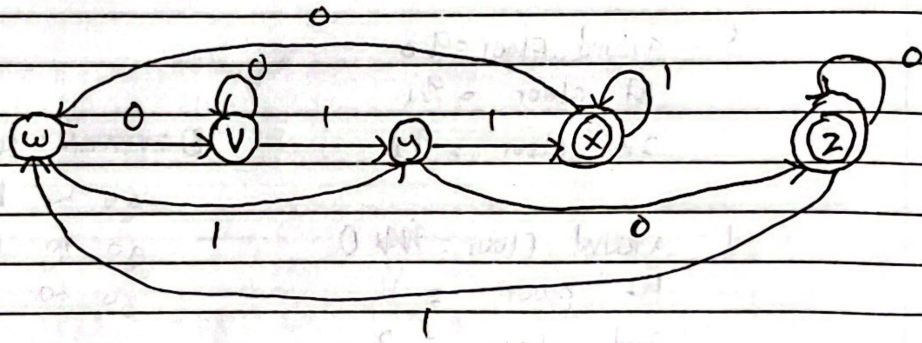


(6) (a) set, $s = \{v, w, x, y, z\}$

initial state = w

(b)



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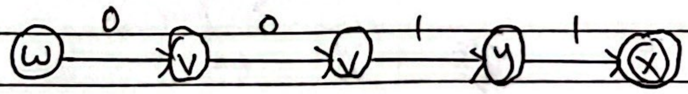

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(7)

$S = \{q_0, q_1, q_2\}$

$I = \{0, 1, 2\}$

$O = \{0, 1, 2, 3\}$

	f_s			f_o		
S	0	1	2	0	1	2
q_0	q_0	q_1	q_2	0	1	3
q_1	q_0	q_1	q_2	1	0	3
q_2	q_0	q_1	q_2	1	2	0

S : ground floor $= q_0$
1st floor $= q_1$
2nd floor $= q_2$

I : ground floor = 0
1st floor = 1
2nd floor = 2

O : remain same floor = 0
go to ~~1st~~ ground floor = 1
go to 1st floor = 2
go to 2nd floor = 3